

Algebra 2 (Advanced)

Unit 6: Solving Cube Root Equations (Day 1)

Multi-Step, Polynomials, & Radicals on Both Sides

Name: _____

I can solve complex cube root equations and those involving rational exponents.

Key Concepts

- **Notation:** $\sqrt[3]{x} = x^{1/3}$. When solving $x^{m/n} = c$, raise to the reciprocal power $\frac{n}{m}$.
- **Polynomials:** If cubing both sides creates a quadratic (x^2) or cubic (x^3), move terms to one side and factor.
- **Even Numerators:** If m is even in $x^{m/n}$, you must consider both positive and negative roots ($\pm$).

Part 1: Multi-Step & Rational Exponents

1. $3(x - 2)^{1/3} = 18$

3. $x^{2/3} = 25$

2. $(x - 2)^{1/3} = 4$

4. $(x - 1)^{4/3} = 16$

Part 2: Radicals on Both Sides

5. $\sqrt[3]{2x - 1} = \sqrt[3]{3x + 5}$

6. $2(5p^2)^{1/3} = (8p + 144)^{1/3}$

Part 3: Solving via Polynomial Factoring

7. $\sqrt[3]{x^2 - 2x} = -1$

8. $(9m^2 - 8m)^{1/3} = m$

$$9. (15x^2 - 7x - 19)^{1/3} = (x^2 - 7x + 6)^{1/3}$$

$$10. \sqrt[3]{x^2 - 16x - 1} = -4$$

Part 4: Applications

11. Tool Wear: A plant models wear (W) in mm vs lifespan (L) in cycles: $\sqrt[3]{L} = 50 - 100W$. If $W = 0.4$ mm, find the maximum operational lifespan (L).

You Try!

$$1. \sqrt[3]{x^2 - 1} = 2$$

$$5. (x - 2)^{1/12} = 1$$

$$2. (x + 10)^{2/3} = 25$$

$$6. x^{7/8} = 128$$

$$3. x^{4/3} = 81$$

$$7. \sqrt[3]{7x^2 - 42x - 1} = (3x + 27)^{1/3}$$

Answer Bank

-22
 -7
 -6
 -4
 -3
 0
 1
 3
 3.4
 4
 5
 6
 8
 9
 15
 27
 66
 115
 125
 218
 256
 1000

4. Metabolic scaling: $\sqrt[3]{L} = 0.1M$. If $M = 15$, find torso length (L).